

Question:

A 67-year-old male presents to his primary care physician complaining of increased fatigue over the last year. He also says that his friends say he appears to be more pale. His past medical history is significant for 10 years of arthritis. Physical exam reveals spoon shaped nails as well as conjunctival pallor. Based on clinical suspicion RBC tests are ordered showing an mean corpuscular volume (MCV) of 75 fl (normal 80-100 fl) and a peripheral blood smear is obtained and found to be normal. Iron studies shows a serum iron of 30 micromolar (normal range 50-170) and a serum ferritin of 300 micrograms/liter (normal range 15-200). What is the most likely diagnosis in this patient? Answer Choices: A. Anemia of chronic disease B. Beta-thalassemia C. Iron deficiency anemia D. Lead poisoning anemia

Reasoning + Answer:

<think> To approach this reasoning problem, we need to analyze the context of the question, which is centered around diagnosing a patient with anemia based on specific lab results and clinical presentations. The key to identifying the correct answer lies in recognizing the context of iron deficiency anemia, as indicated by the significantly low serum iron and ferritin levels, despite a normal peripheral blood smear and RBC MCV. This directly points to option C, iron deficiency anemia. </think> <answer> Iron deficiency anemia </answer>

